NETCORE SYSTEMS LIMITED

NETWORK DESIGN, CONFIGURATION, AND SECURITY REPORT

Cisco Packet Tracer Case Study

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1. **Project Overview**

Netcore Systems Limited required a structured and secure network to improve communication between IT, Sales and HR departments. The network was designed and configured in Cisco Packet Tracer using a router, departmental switches, end user devices, and a central server. The project also included IP addressing, DHCP, DNS, HTTP services connectivity testing, and an Access Control List (ACL) to restrict unauthorized communication between departments.

2. **Network Requirements and Objectives**

   The main requirements and objectives of the project were to:

   - Design a structured network for IT, Sales and HR departments.
   - Configure a router and departmental switches to enable communication between the networks.
   - Assign appropriate IP addresses to each department.
   - Configure centralized DHCP services to automatically assign IP addresses to end-user devices.
   - Configure DNS and HTTP services on the central server.
   - Test connectivity between devices and departments.
   - Apply an Access Control List (ACL) to restrict unauthorized communication between departments.
   - Verify that the completed network operates correctly and securely.

3. **Network Topology and Design**

   The Network was designed in Cisco Packet Tracer using one Cisco 2911 router, three Cisco 2960 switches, end-user devices, and a central server. Each department was placed on a separate network to improve organization, communication, and control. The IT department hosted the central server, while the router provided communication between the IT, Sales, and HR networks.

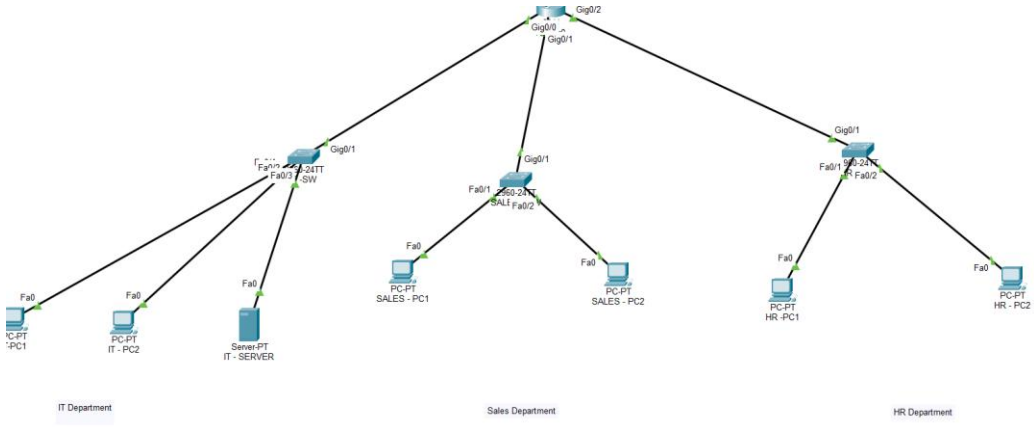


Figure 1: Netcore Systems Limited Network Topology

Figure 1 shows the completed topology for Netcore Systems Limited. The Cisco 2911 router connects the IT, Sales, and HR departmental networks through their respective Cisco 2960 switches. The IT network also contains the central server, which provides network services to the connected devices.

4. IP Addressing Scheme

A separate IP network was assigned to each department to improve organization, routing, and network management. The addressing scheme also defined the default gateway and server addresses used in the network.

Department	Network	Subnet Mask	Default Gateway
IT	192.168.10.0	255.255.255.0	192.168.10.1
Sales	192.168.20.0	255.255.255.0	192.168.20.1
HR	192.168.30.0	255.255.255.0	192.168.30.1

5. **Router Configuration**

The Cisco 2911 router was configured to provide communication between the IT, Sales, and HR networks. Each router interface was assigned an IP address that served as the default gateway for its respective department.

- GigabitEthernet0/0 - 192.168.10.1 - IT Network
- GigabitEthernet0/1 - 192.168.20.1 - Sales Network
- GigabitEthernet0/2 - 192.168.30.1 - HR Network

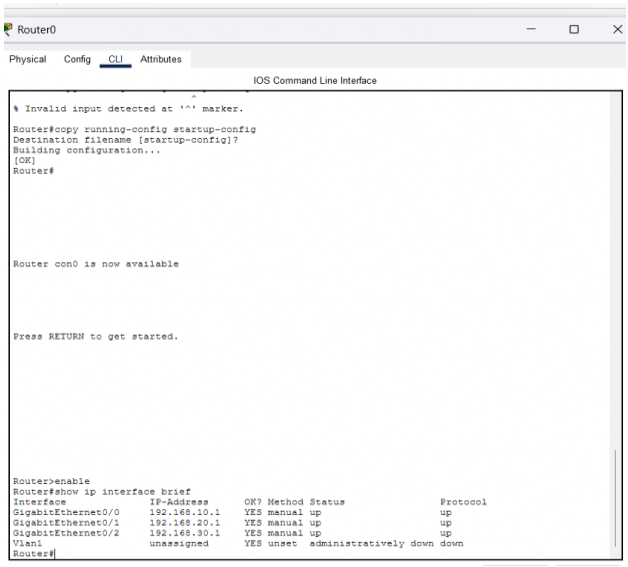


Figure 2: Router Interface IP Configuration

Figure 2 shows the router interface configuration verified using the show ip interface brief command. The output confirms that GigabitEthernet0/0, GigabitEthernet0/1, and

GigabitEthernet0/2 were assigned the correct IP addresses for the IT, Sales, and HR networks, and the interfaces were active.

6. DHCP Configuration

A centralized DHCP service was configured on the server to automatically assign IP addresses to devices in IT, Sales, and HR departments. Separate DHCP pools were created for each network using the appropriate default gateway, DNS server, subnet mask, and starting IP address.

- IT Pool: Gateway 192.168.10.1 | Start IP 192.168.10.5 | Subnet Mask 255.255.255.0
- Sales Pool: Gateway 192.168.20.1 | Start IP 192.168.20.5 | Subnet Mask 255.255.255.0
- HR Pool: Gateway 192.168.30.1 | Start IP 192.168.30.5 | Subnet Mask 255.255. 255.0
- DNS Server: 192.168.10.2

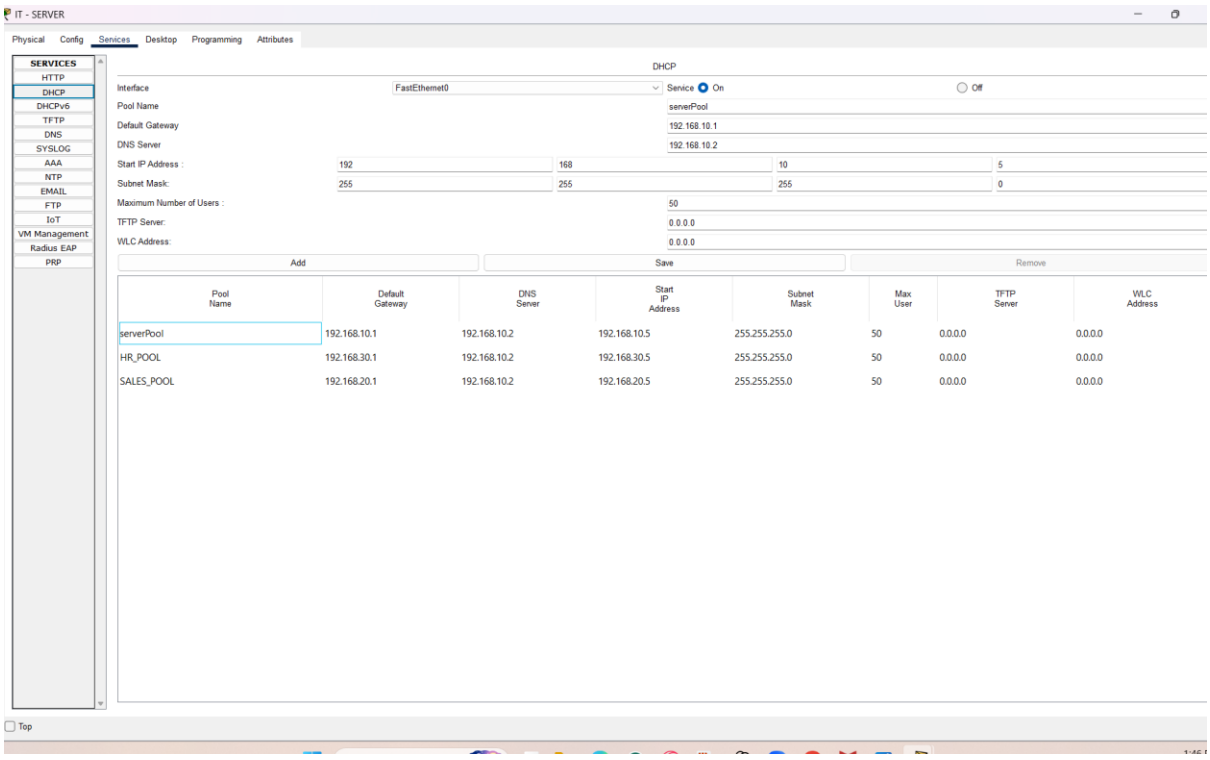


Figure 3: DHCP Server Configuration

Figure 3 shows the centralized DHCP configuration on the server. Separate pools were created for the IT, Sales, and HR networks, with the correct default gateways, DNS server, starting IP addresses, and subnet mask.

7. DNS and HTTP Services

DNS and HTTP services were configured on the central server to support name resolution and web access across the network. The DNS service was used to map the configured website name to the server IP address, while the HTTP service hosted the Netcore Systems welcome webpage.

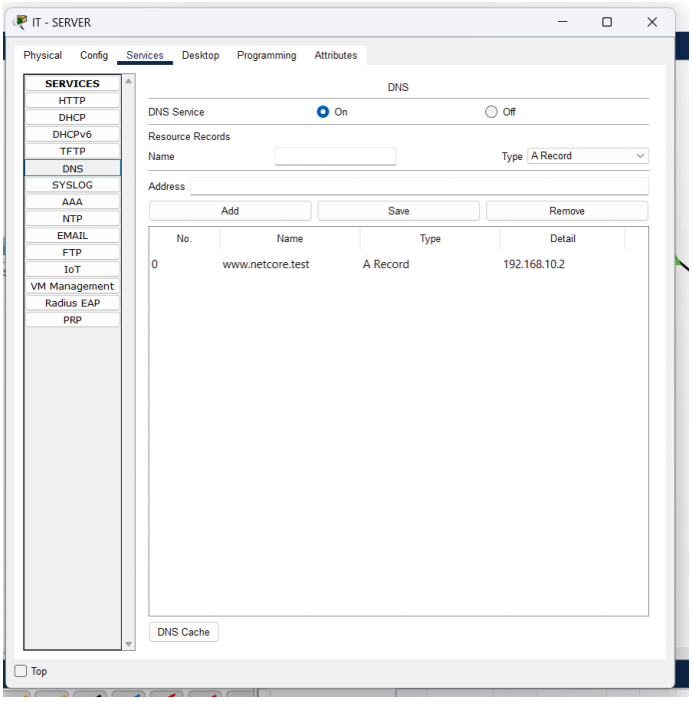


Figure 4: DNS Server Configuration

Figure 4 shows the DNS configuration on the central server. An A record was created for www.netcore.test and mapped to the server IP address 192.168.10.2, allowing devices on the network to resolve the website name to server.

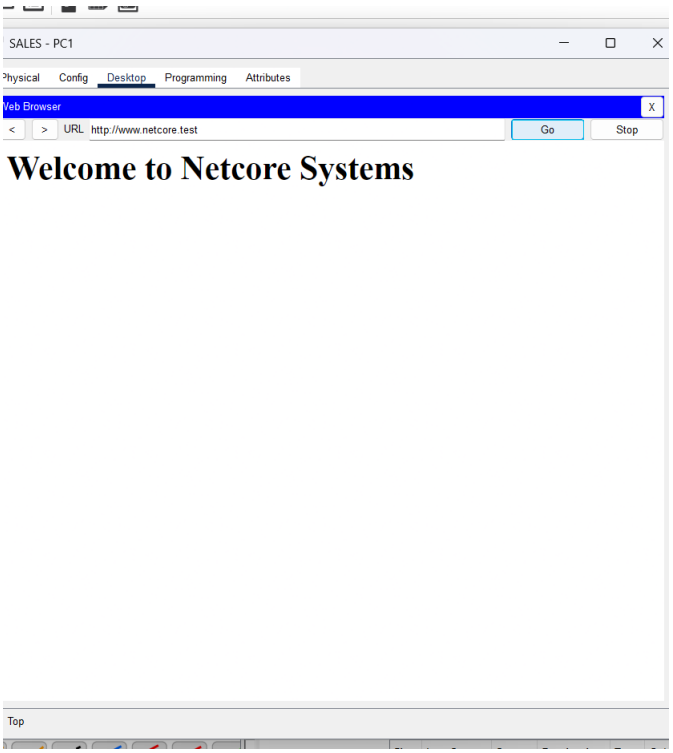


Figure 5: HTTP Web Service Verification

Figure 5 shows successful access to the Netcore Systems webpage from a Sales PC, confirming that the HTTP service on the central server was reachable across the network.

8. **Access Control List (ACL) Configuration**

An Access Control List (ACL) was configured on the router to improve network security by controlling communication between departments. The ACL was designed to block ICMP traffic from the Sales network to the IT network while allowing other required network communication.

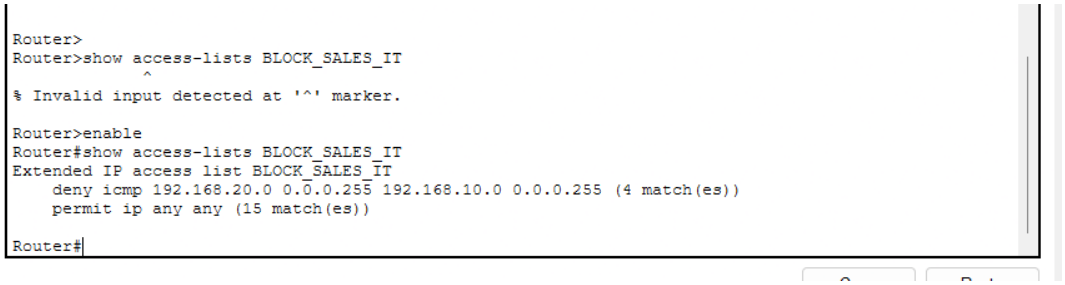


Figure 6: ACL Configuration

Figure 6 shows the BLOCK_SALES_IT access control list configured on the router. The ACL denies ICMP traffic from the Sales network 192.168.20.0/24 to the IT network 192.168.10.0/24 while permitting other IP traffic.



Figure 7: ACL Interface Application

Figure 7 shows the BLOCK_SALES_IT ACL applied inbound on Sales-facing GigabitEthernet0/1 interface, so Sales traffic is checked as it enters the router.

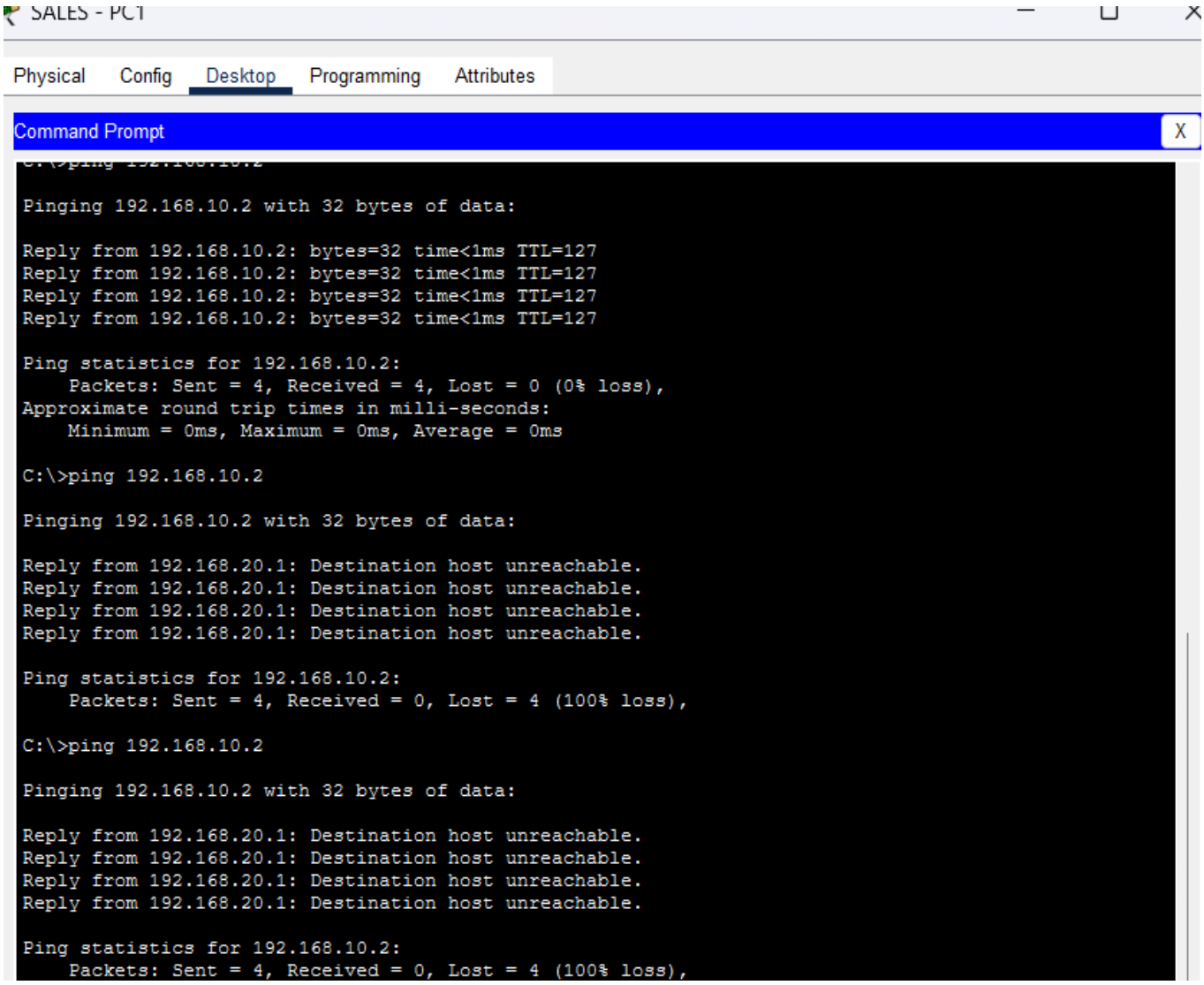


Figure 8: ACL Block Test

Figure 8 shows the connectivity test from the Sales network to the IT network after the ACL was applied. The failed ICMP ping confirms that the ACL successfully blocked the restricted traffic as intended.

9. Connectivity and validation Testing

Connectivity testing was performed to verify that device could communicate across the configured departmental networks before the security restriction was applied. Ping tests and web access were used to confirm routing, server reachability, and network services. The ACL was then tested separately to confirm that the required Sales to-IT ICMP traffic was blocked.

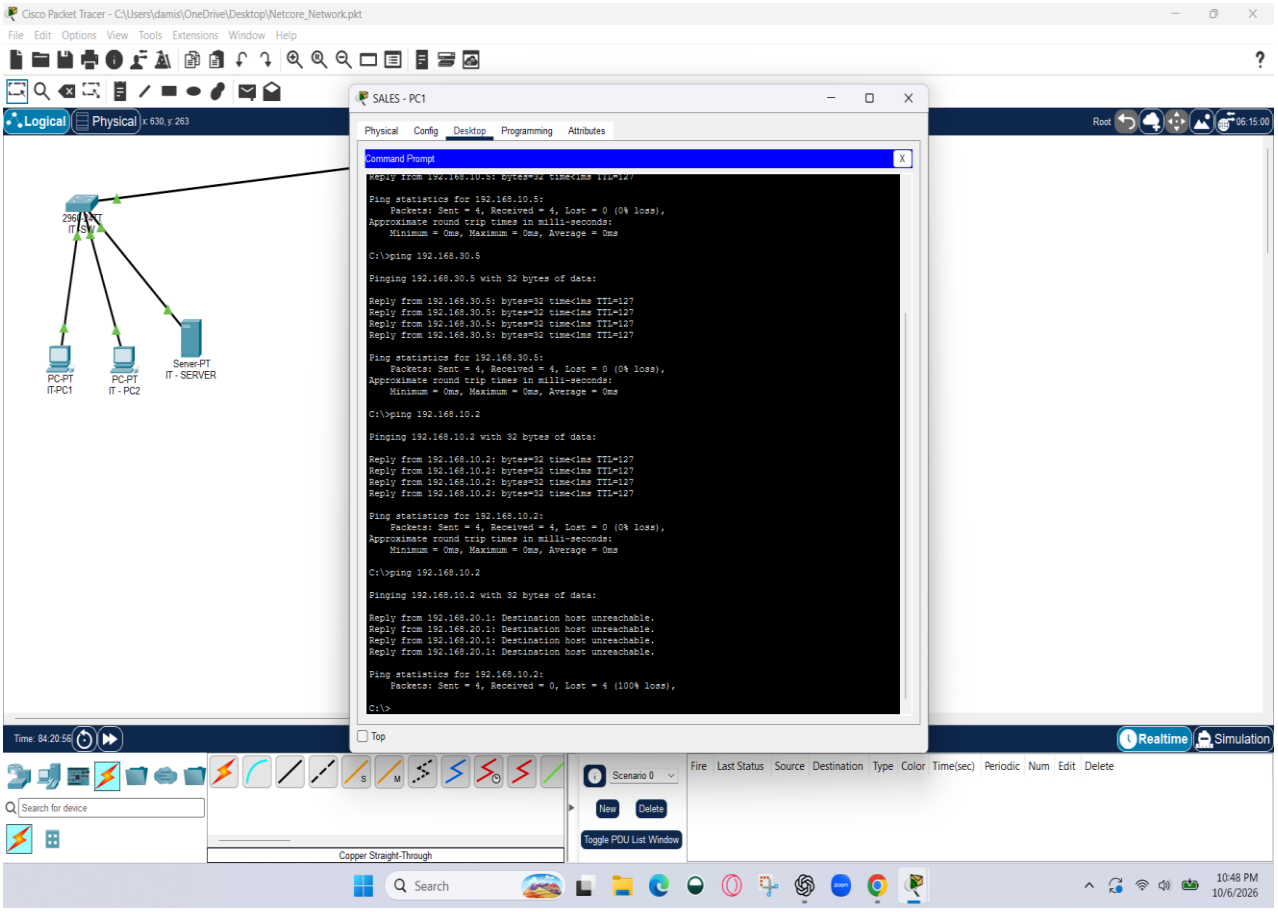


Figure 9: Connectivity Test from Sales PC

Figure 9 provides evidence of the connectivity tests performed from the Sales PC.

10. Challenges and Lessons Learned

One **c**hallenge during the project was ensuring the router interfaces, DHCP pools, DNS service, and ACL settings all worked together correctly. Testing helped identify configuration issues and confirm that each service was functioning as expected.

The project improved my understanding of IP addressing, routing, DHCP, DNS, HTTP services, connectivity testing, and the use of Access Control Lists to enforce network security. I also learned the importance of testing each configuration step and documenting evidence clearly.

11. Conclusion

The Netcore Systems Limited network was successfully designed, configured, secured, and tested in Cisco Packet Tracer. The completed network provided communication between the IT, Sales, and HR departments, centralized network services, and controlled access through an ACL. Testing confirmed that the required network services were operational and that the security restriction worked as intended.